Nature versus Artefacts: Places and Objects in Nineteenth-Century Novels from Spain and Latin America

Henny-Krahmer, Ulrike
ulrike.henny-krahmer@uni-rostock.de
Universität Rostock, Germany

Müller, Caroline
caroline.mueller4@uni-rostock.de
Universität Rostock, Germany

Introduction

Space, places, and spatial objects have been of interest for literary historical research for a long time, since narrative space is one of the central components of narratological analysis. The general importance for place in literary texts has been emphasized (Lutwack 1984) but there has also been a focus on the relevance of spatial concepts for literary historical periods such as Romanticism or Realism in the nineteenth century (Pape 2009, Mucignat 2013). In addition, narrative space has been analysed in the field of Computational Literary Studies (CLS), where narratological concepts of space have been approached by operationalisable textual features (Schumacher 2023; Henny-Krahmer et al. 2024a, 2024b).

Regarding the semantics of narrative space, the contrast between natural places and those created by men, i.e., artificial places or artefacts, is a recurring topic in the research about place in literature, even in computational studies (Grisot and Herrmann 2023) and in the context of eco-critical research (Hall 2014). With this contribution, we take up this state of research by analysing natural and artificial spaces, places, and spatial objects in nineteenth-century novels from Spain and Latin America with a computational approach, thereby producing new research results in the CLS for Spanish texts. We aim to address the following questions: Which kind of places and spatial objects are mentioned more frequently in the texts, natural ones or artefacts? Are there differences between texts from Spain and Latin America? Are there differences between texts from different literary currents? This poster contribution reflects the current state of our ongoing work.

Data and Annotation

The overall corpora that we aim to analyse consist of 200 texts, 100 from Spain and 100 from Latin America. The sources of these corpora are ELTeC-spa (Navarro Colorado 2021) and Conha19 (Henny-Krahmer 2021). The texts were published 1840–1920, are chosen to represent ranges of 20 years in a balanced way, and include texts of many different authors. We annotated 20 randomly chosen chapters of 20 novels manually with a category set for natural and artificial places and objects that we defined using the annotation tool CATMA (Gius et al. 2024). There are six categories (CONTAINER-ARTEFACT, CONTAINER-NATURAL, CONTAINER-REGION, CONTAINER-SETTLEMENT, OBJECT-ARTEFACT, OBJECT-NATURAL) exemplified in the sentences in Fig. 1.

(1) "En todos los [distritos]$_{\text{CONTAINER-REGION}}$ un [café]$_{\text{CONTAINER-ARTEFACT}}$ y una [taberna]$_{\text{CONTAINER-ARTEFACT}}$ desobedecían tranquilamente las órdenes de la autoridad."

(2) "¿a quién se le ocurre dar en invierno y por la noche, una fiesta en una [estancia]$_{\text{CONTAINER-ARTEFACT}}$ de [campo]$_{\text{CONTAINER-NATURAL}}$?"

(3) "En un [divan]$_{\text{OBJECT-ARTEFACT}}$, frente á la [chimenea]$_{\text{OBJECT-ARTEFACT}}$, estaba reclinada la marquesa […]."

(4) "Una noche en que la [luna]$_{\text{OBJECT-NATURAL}}$ empezaba a menguar, y que algunas [nubes]$_{\text{OBJECT-NATURAL}}$ cubrían el [cielo]$_{\text{CONTAINER-NATURAL}}$, estaban los dos sentados en una [peña]$_{\text{OBJECT-NATURAL}}$ […]"

Figure 1: Example sentences with annotation categories for natural places and artefacts.

Two people annotated the 20 chapters and a gold standard annotation was derived from the two annotation layers. The resulting ground truth that forms the basis for training a corresponding model is shown in Fig. 2.

Entity	# All	# Train	# Test	# Devel
CONTAINER-ARTEFACT	788	623	91	74
CONTAINER-NATURAL	316	247	29	40
CONTAINER-REGION	116	98	6	12
CONTAINER-SETTLEMENT	129	98	19	12
OBJECT-ARTEFACT	316	259	33	24
OBJECT-NATURAL	169	129	19	21
unlabeled words	78269	62168	8321	7780

Figure 2: Ground truth for natural places and artefacts in Spanish texts.

A first observation that we can make from the manually annotated data is that artificial containers and objects are about two times more frequent than natural containers and objects, which is a finding that we did not expect beforehand.

Model Training and Analysis

The tool NTEE (Sperfeld et al. 2023) was used to train a model based on a Spanish language model (bert-base-es-cased).[1] We trained 10 epochs, reaching 0.748 as the highest entity-wise F1 score on the validation set (see the training course in Fig. 3).

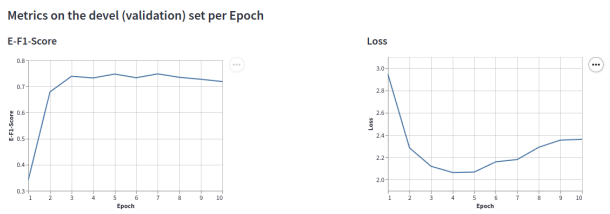

Figure 3: Training course of the model for recognizing natural places and artefacts.

So far, we have used the model to predict the references to places and objects in 195 texts.[2] The top five artefacts and natural containers and objects mentioned in the texts are shown in Fig. 4.

category	rank	token	occurrences
artefact	1	casa	20,916
	2	puerta	9,235
	3	calle	5,657
	4	mesa	3,764
	5	cuarto	3,281
natural container/object	1	cielo	4,668
	2	luz	4,638
	3	sol	4,005
	4	aire	2,620
	5	tierra	2,487

Figure 4: Top five artefacts and natural containers and objects.

No significant differences could be found so far between the sub-corpora from Spain and Latin America. Looking at the literary movements, it can be stated that spatial references are more frequent overall in the peak phase of Realism and Naturalism (around 1880–1900) than before and after.

Conclusions and Next Steps

By developing guidelines for annotating references to natural places and objects versus artefacts and using them for the annotation of Spanish texts, we found that artificial, man-made places and objects are mentioned more frequently in the novels than natural ones, a result that we did not expect from the outset. It was possible to train a model for recognizing these references to space with an F1 score of around 0.75. We still have to analyse the kinds of mistakes that are made by the model to see if we can improve the annotations and model training in the future. Furthermore, we aim to predict spatial references for all of the 200 novels of the corpus and to conduct further analyses regarding the questions outlined in the introduction.

Notes

1. .
2. For the five remaining texts, format problems with the predictions still need to be solved.

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